

CLASS 11 MATHEMATICS

COMPLEX NUMBERS

&

QUADRATIC EQUATIONS

SHORT REVISION NOTES

Learn the Concept → Remember the Formula → Solve Smartly

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Mathematics

1. The Imaginary Unit i

The imaginary unit is defined as

$$i = \sqrt{-1}$$

Therefore,

$$i^2 = -1$$

The powers of i repeat after every four powers:

$$i, -1, -i, 1, i, -1, -i, 1, \dots$$

Quick Trick

To find i^n , divide n by 4.

Remainder	i^n
0	1
1	i
2	-1
3	$-i$

Example:

$$i^{27} = i^{4(6)+3} = i^3 = -i$$

2. Complex Number

A number of the form

$$z = a + ib$$

is called a complex number, where $a, b \in \mathbb{R}$.

Here,

$$\operatorname{Re}(z) = a$$

and

$$\operatorname{Im}(z) = b$$

Remember

If

$$z = 5 - 7i,$$

then

$$\operatorname{Re}(z) = 5, \quad \operatorname{Im}(z) = -7.$$

The imaginary part is -7 , not $-7i$.

3. Types of Complex Numbers

For

$$z = a + ib$$

we have:

- If $b = 0$, then z is a **real number**.
- If $a = 0$, then $z = ib$ is a **purely imaginary number**.
- If $a \neq 0$ and $b \neq 0$, then z is a **non-real complex number**.

$$\mathbb{R} \subset \mathbb{C}$$

Every real number is a complex number with imaginary part zero.

4. Equality of Complex Numbers

If

$$a + ib = c + id,$$

then

$$a = c, \quad b = d$$

Example:

$$3x + i(2y) = 6 + 5i$$

Comparing real and imaginary parts,

$$3x = 6, \quad 2y = 5.$$

Hence,

$$x = 2, \quad y = \frac{5}{2}$$

5. Addition and Subtraction

Let

$$z_1 = a + ib, \quad z_2 = c + id.$$

Then

$$z_1 + z_2 = (a + c) + i(b + d)$$

and

$$z_1 - z_2 = (a - c) + i(b - d)$$

Example:

$$(3 + 2i) + (5 - 7i)$$

$$= 8 - 5i$$

Therefore,

$$8 - 5i$$

6. Multiplication

$$(a + ib)(c + id)$$

Expanding,

$$\begin{aligned}(a + ib)(c + id) &= ac + iad + ibc + i^2bd \\ &= ac + iad + ibc - bd.\end{aligned}$$

Therefore,

$$(a + ib)(c + id) = (ac - bd) + i(ad + bc)$$

Must Remember

Always use

$$i^2 = -1$$

while multiplying complex numbers.

7. Conjugate

If

$$z = a + ib,$$

then its conjugate is

$$\bar{z} = a - ib$$

Important identities:

$$z + \bar{z} = 2 \operatorname{Re}(z)$$

$$z - \bar{z} = 2i \operatorname{Im}(z)$$

and

$$z\bar{z} = a^2 + b^2$$

8. Modulus

For

$$z = a + ib,$$

the modulus of z is

$$|z| = \sqrt{a^2 + b^2}$$

Also,

$$|z|^2 = z\bar{z}$$

Example:

If

$$z = 3 + 4i,$$

then

$$|z| = \sqrt{3^2 + 4^2} = \sqrt{25} = \boxed{5}.$$

Important Properties

$$|z| \geq 0$$

$$|z| = 0 \iff z = 0$$

$$|\bar{z}| = |z|$$

$$|z_1 z_2| = |z_1| |z_2|$$

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|} \quad (z_2 \neq 0)$$

9. Division of Complex Numbers

To divide complex numbers, multiply numerator and denominator by the conjugate of the denominator. Consider

$$\frac{a + ib}{c + id}.$$

Multiply by

$$\frac{c - id}{c - id}.$$

Therefore,

$$\frac{a + ib}{c + id} = \frac{(a + ib)(c - id)}{c^2 + d^2}$$

Golden Rule

Denominator ko real banao using its conjugate.

10. Argand Plane

A complex number

$$z = a + ib$$

can be represented by the point

$$(a, b)$$

in the Cartesian plane.

The horizontal axis is called the **real axis**.The vertical axis is called the **imaginary axis**.

Thus,

$$z = a + ib \longleftrightarrow (a, b)$$

Also,

$$|z| = \text{distance of } z \text{ from the origin}$$

11. Polar Form

Let

$$z = a + ib.$$

Put

$$r = |z| = \sqrt{a^2 + b^2}.$$

If θ is the argument of z , then

$$a = r \cos \theta, \quad b = r \sin \theta.$$

Hence,

$$z = r(\cos \theta + i \sin \theta)$$

This is called the **polar form** of a complex number.

12. Argument

For

$$z = a + ib,$$

the argument is denoted by

$$\arg z = \theta.$$

We have

$$\tan \theta = \frac{b}{a}$$

provided $a \neq 0$.

Important Warning

Do not determine the argument only from $\tan \theta$.
Always check the quadrant using the signs of a and b .

13. Quadratic Equation

A quadratic equation is an equation of the form

$$ax^2 + bx + c = 0, \quad a \neq 0$$

where $a, b, c \in \mathbb{R}$.

Here,

a = coefficient of x^2 ,

b = coefficient of x ,

c = constant term.

14. Quadratic Formula

For

$$ax^2 + bx + c = 0,$$

the roots are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The quantity

$$\Delta = b^2 - 4ac$$

is called the **discriminant**.

15. Nature of Roots

The discriminant tells us the nature of the roots.

Condition	Nature of Roots
$\Delta > 0$	Two distinct real roots
$\Delta = 0$	Two equal real roots
$\Delta < 0$	Non-real complex roots

Memory Trick

$$\begin{aligned} \Delta > 0 &\Rightarrow \text{Real and Different} \\ \Delta = 0 &\Rightarrow \text{Real and Same} \\ \Delta < 0 &\Rightarrow \text{Complex} \end{aligned}$$

16. Relation Between Roots and Coefficients

Suppose α and β are the roots of

$$ax^2 + bx + c = 0.$$

Then

$$\alpha + \beta = -\frac{b}{a}$$

and

$$\alpha\beta = \frac{c}{a}$$

These two formulas are extremely important.

Useful Results

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$$

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$$

17. Formation of Quadratic Equation

If α and β are the roots, then

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

is the required quadratic equation.

Example:

Let the roots be 3 and -5 .

Then

$$\alpha + \beta = 3 - 5 = -2$$

and

$$\alpha\beta = -15.$$

Therefore,

$$x^2 - (-2)x - 15 = 0$$

or

$$x^2 + 2x - 15 = 0$$

18. Complex Roots of a Quadratic

For a quadratic equation with real coefficients,

$$ax^2 + bx + c = 0,$$

if

$$\Delta < 0,$$

then the roots are non-real complex numbers.

The roots occur in conjugate pairs.

If one root is

$$\alpha + i\beta,$$

then the other root is

$$\boxed{\alpha - i\beta}.$$

Key Result

Complex roots of a quadratic equation with real coefficients occur in conjugate pairs.

19. One-Page Formula Sheet

COMPLEX NUMBERS

$$\boxed{i^2 = -1}$$

$$\boxed{z = a + ib}$$

$$\boxed{\operatorname{Re}(z) = a}$$

$$\boxed{\operatorname{Im}(z) = b}$$

$$\boxed{\bar{z} = a - ib}$$

$$\boxed{|z| = \sqrt{a^2 + b^2}}$$

$$\boxed{z\bar{z} = |z|^2}$$

$$\boxed{z = r(\cos \theta + i \sin \theta)}$$

$$\boxed{\tan \theta = \frac{b}{a}}$$

QUADRATIC EQUATIONS

$$ax^2 + bx + c = 0$$

$$\Delta = b^2 - 4ac$$

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}$$

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

20. Last-Minute Exam Checklist

1. Remember the four-power cycle of i .
2. Always write a complex number in the form $a + ib$.
3. Do not confuse $\text{Im}(z)$ with ib .
4. For division, multiply by the conjugate.
5. Remember $|z|^2 = z\bar{z}$.
6. In a quadratic equation, calculate the discriminant first.
7. $\Delta > 0$ gives two distinct real roots.
8. $\Delta = 0$ gives equal real roots.
9. $\Delta < 0$ gives complex conjugate roots.
10. Remember $\alpha + \beta = -b/a$.
11. Remember $\alpha\beta = c/a$.
12. To form a quadratic from roots, use $x^2 - (\alpha + \beta)x + \alpha\beta = 0$.

CONCEPT → FORMULA → PRACTICE → MASTERY

Don't just memorize mathematics — understand why it works!

“‘latex

21. Important Questions with Solutions

Q1. Complex Numbers

Question

If

$$(x + iy)^3 = u + iv,$$

show that

$$\frac{u}{x} + \frac{v}{y} = 4(x^2 - y^2)$$

Solution:

We have

$$(x + iy)^3 = u + iv.$$

Using

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3,$$

we get

$$\begin{aligned}(x + iy)^3 &= x^3 + 3x^2(iy) + 3x(iy)^2 + (iy)^3 \\ &= x^3 + 3x^2iy - 3xy^2 - iy^3 \\ &= (x^3 - 3xy^2) + i(3x^2y - y^3).\end{aligned}$$

Comparing real and imaginary parts,

$$u = x^3 - 3xy^2$$

and

$$v = 3x^2y - y^3.$$

Therefore,

$$\frac{u}{x} = x^2 - 3y^2$$

and

$$\frac{v}{y} = 3x^2 - y^2.$$

Adding,

$$\begin{aligned}\frac{u}{x} + \frac{v}{y} &= x^2 - 3y^2 + 3x^2 - y^2 \\ &= 4x^2 - 4y^2 \\ &= 4(x^2 - y^2).\end{aligned}$$

Hence proved.

Key Idea: Expand the complex expression and compare real and imaginary parts.

Q2. Sequence and Series

Question

Find the sum of the sequence

$$7, 77, 777, \dots$$

to n terms.

Solution:

The k th term is

$$7, 77, 777, \dots$$

Observe that

$$7 = \frac{7}{9}(10 - 1),$$

$$77 = \frac{7}{9}(10^2 - 1),$$

$$777 = \frac{7}{9}(10^3 - 1).$$

Hence,

$$T_k = \frac{7}{9}(10^k - 1)$$

Therefore,

$$S_n = \sum_{k=1}^n \frac{7}{9}(10^k - 1).$$

Thus,

$$S_n = \frac{7}{9} \left(\sum_{k=1}^n 10^k - n \right).$$

Using

$$\sum_{k=1}^n 10^k = \frac{10(10^n - 1)}{9},$$

we get

$$S_n = \frac{7}{9} \left[\frac{10(10^n - 1)}{9} - n \right].$$

Hence,

$$S_n = \frac{7}{81} [10(10^n - 1) - 9n]$$

or

$$S_n = \frac{7}{81}(10^{n+1} - 10 - 9n).$$

Key Idea:

$$7, 77, 777, \dots = \frac{7}{9}(10^k - 1)$$

This converts the problem into a geometric progression.

Q3. Complex Numbers

Question

Reduce

$$\left(\frac{1}{1-4i} - \frac{2}{1+i} \right)$$

to the standard form $a + ib$.

Also find its modulus and conjugate.

Solution:

First,

$$\frac{1}{1-4i} = \frac{1+4i}{(1-4i)(1+4i)}.$$

Therefore,

$$\frac{1}{1-4i} = \frac{1+4i}{17}.$$

Similarly,

$$\frac{2}{1+i} = \frac{2(1-i)}{(1+i)(1-i)} = \frac{2(1-i)}{2} = 1-i.$$

Hence,

$$\begin{aligned} z &= \frac{1+4i}{17} - (1-i) \\ &= \frac{1+4i}{17} - 1 + i \\ &= -\frac{16}{17} + \frac{21}{17}i. \end{aligned}$$

Therefore, the standard form is

$$z = -\frac{16}{17} + \frac{21}{17}i$$

Modulus

For

$$z = a + ib,$$

$$|z| = \sqrt{a^2 + b^2}.$$

Therefore,

$$\begin{aligned}|z| &= \sqrt{\left(-\frac{16}{17}\right)^2 + \left(\frac{21}{17}\right)^2} \\&= \frac{1}{17}\sqrt{256 + 441} \\&= \frac{\sqrt{697}}{17}.\end{aligned}$$

Hence,

$$|z| = \frac{\sqrt{697}}{17}$$

Conjugate

The conjugate of

$$z = -\frac{16}{17} + \frac{21}{17}i$$

is

$$\bar{z} = -\frac{16}{17} - \frac{21}{17}i$$

Final Answer:

$$z = -\frac{16}{17} + \frac{21}{17}i$$

$$|z| = \frac{\sqrt{697}}{17}$$

$$\bar{z} = -\frac{16}{17} - \frac{21}{17}i$$

22. Important Quadratic Equation Questions

Q4. Roots of a Quadratic Equation

Question

Find the roots of

$$2x^2 - 5x + 3 = 0.$$

Solution:

Here,

$$a = 2, \quad b = -5, \quad c = 3.$$

Using

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a},$$

we get

$$\begin{aligned} x &= \frac{5 \pm \sqrt{25 - 24}}{4} \\ &= \frac{5 \pm 1}{4}. \end{aligned}$$

Therefore,

$$x = \frac{6}{4} = \frac{3}{2}$$

or

$$x = \frac{4}{4} = 1.$$

Hence,

$$\boxed{x = 1, \frac{3}{2}}.$$

Q5. Nature of Roots

Question

Determine the nature of the roots of

$$3x^2 - 4x + 5 = 0.$$

Solution:

Here,

$$a = 3, \quad b = -4, \quad c = 5.$$

The discriminant is

$$\begin{aligned}\Delta &= b^2 - 4ac \\ &= (-4)^2 - 4(3)(5) \\ &= 16 - 60 \\ &= -44.\end{aligned}$$

Since

$$\Delta < 0,$$

the equation has

two non-real complex conjugate roots.

Q6. Relation Between Roots

Question

If α and β are the roots of

$$x^2 - 7x + 10 = 0,$$

find

$$\alpha^2 + \beta^2.$$

Solution:

Comparing with

$$ax^2 + bx + c = 0,$$

we have

$$\alpha + \beta = 7$$

and

$$\alpha\beta = 10.$$

Now,

$$\begin{aligned}\alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\ &= 7^2 - 2(10) \\ &= 49 - 20 \\ &= \boxed{29}.\end{aligned}$$

Q7. Equal Roots

Question

Find the value of k for which

$$x^2 + kx + 9 = 0$$

has equal roots.

Solution:

For equal roots,

$$\Delta = 0.$$

Therefore,

$$k^2 - 4(1)(9) = 0.$$

Thus,

$$k^2 - 36 = 0$$

and

$$(k - 6)(k + 6) = 0.$$

Hence,

$$\boxed{k = \pm 6}.$$

Q8. Complex Roots**Question**

Solve

$$x^2 + 4x + 13 = 0$$

and express the roots in the form $a + ib$.

Solution:

Here,

$$a = 1, \quad b = 4, \quad c = 13.$$

The discriminant is

$$\Delta = 4^2 - 4(1)(13) = 16 - 52 = -36.$$

Using the quadratic formula,

$$\begin{aligned} x &= \frac{-4 \pm \sqrt{-36}}{2} \\ &= \frac{-4 \pm 6i}{2}. \end{aligned}$$

Therefore,

$$\boxed{x = -2 + 3i}$$

or

$$\boxed{x = -2 - 3i}.$$

Thus the roots are complex conjugates. ““